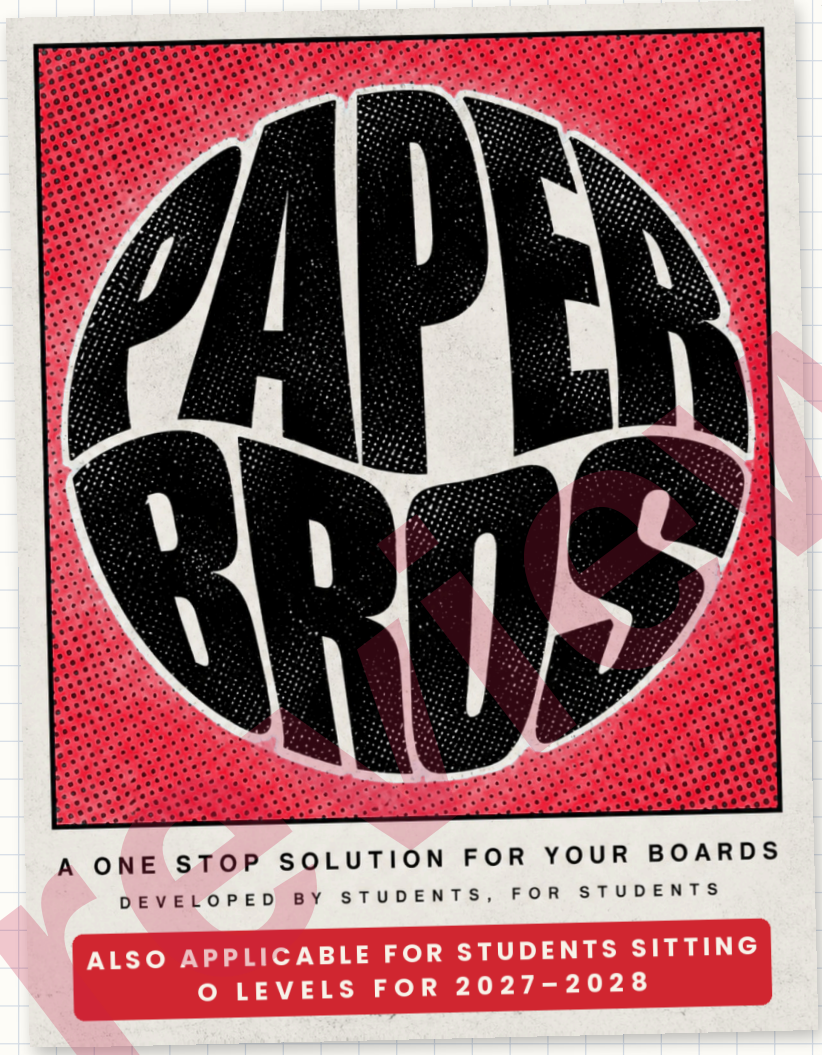
PREVIEW

Cambridge O Level

Chemistry 5070

the complete notes – for exams in 2029



notes for every syllabus point · worked examples
accurate diagrams & graphs · Paper 1 & Paper 2 practice + answers

this notebook belongs to:

Name .....

School / class ..... Exam: June / Nov

Preview

PREVIEW

what's inside

Contents

tick the box when a topic is done & dusted ✓

#	Topic	What's inside (syllabus sections)	Notes pg	Practice pg	✓
–	The exam at a glance	papers, marks, AOs, command words, how to use these notes	3	–	☐
1	States of matter	1.1 solids, liquids & gases, changes of state, heating/cooling curves, gas volume · 1.2 diffusion	4	6	☐
2	Atoms, elements & compounds	2.1 elements, compounds, mixtures · 2.2 atomic structure · 2.3 isotopes · 2.4 ionic bonding · 2.5 covalent bonding · 2.6 giant covalent · 2.7 metallic bonding	8	13	☐
3	Stoichiometry	3.1 formulas & equations · 3.2 A_r and M_r · 3.3 the mole, gas volumes, concentration, titration, empirical formula, % yield & purity	15	19	☐
4	Electrochemistry	4.1 electrolysis, half-equations, electroplating · 4.2 hydrogen–oxygen fuel cells	21	24	☐
5	Chemical energetics	5.1 exo/endothermic, ΔH, activation energy, pathway diagrams, bond energy calculations	26	28	☐
6	Chemical reactions	6.1 physical/chemical change · 6.2 rate of reaction · 6.3 reversible reactions, equilibrium, Haber & Contact · 6.4 redox	30	35	☐
7	Acids, bases & salts	7.1 properties of acids & bases, indicators, pH, strong/weak · 7.2 oxides · 7.3 preparing salts, solubility rules	37	41	☐
8	The Periodic Table	8.1 arrangement & trends · 8.2 Group I · 8.3 Group VII · 8.4 transition elements · 8.5 noble gases	43	47	☐
9	Metals	9.1 properties · 9.2 uses · 9.3 alloys · 9.4 reactivity series · 9.5 corrosion · 9.6 extraction (iron & aluminium)	49	55	☐
10	Chemistry of the environment	10.1 water · 10.2 fertilisers · 10.3 air quality & climate	58	62	☐
11	Organic chemistry	11.1 formulas & terms · 11.2 naming · 11.3 fuels · 11.4 alkanes · 11.5 alkenes · 11.6 alcohols · 11.7 carboxylic acids · 11.8 polymers	64	71	☐
12	Experimental techniques & chemical analysis	12.1 apparatus & terms · 12.2 titrations · 12.3 chromatography · 12.4 separation & purification · 12.5 tests for ions & gases · practical-paper skills	74	79	☐
✶	Answers	Paper 1 keys + Paper 2 mark-scheme style answers for every topic	82	–	☐

Same content as 2027 & 2028 ✔

Cambridge says the 2029 syllabus has **no significant changes which affect teaching** – so these notes work for the June/Nov 2027 and 2028 series too. Textbooks endorsed from 2023 are still fine.

How I'd use this book

1. read the notes → 2. cover them & redo the worked examples → 3. do the practice without peeking → 4. mark with the answers at the back & fix your mistakes in red.

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before we start...

The Exam at a Glance

You sit **three** papers. Everyone takes Paper 1 **and** Paper 2, then **either** Paper 3 **or** Paper 4. Grades go from **A*** down to **E**.

Paper	Time	Marks	Weight	What it looks like	Tests
1 Multiple Choice	1 h	40	30%	40 compulsory four-option (A–D) questions	AO1 + AO2
2 Theory	1 h 45 min	80	50%	compulsory short-answer & structured questions	AO1 + AO2
3 Practical Test	1 h 30 min	40	20%	you actually do experiments in a lab	AO3
		– OR –			
4 Alternative to Practical	1 h	40	20%	written questions about experiments (no lab)	AO3

AO Assessment objectives

Objective	Weight
AO1 Knowledge with understanding – recall facts, definitions, equations, safety, uses	50%
AO2 Handling information & problem-solving – data, graphs, calculations, unfamiliar contexts	30%
AO3 Experimental skills & investigations – planning, measuring, evaluating	20%

Inside Papers 1 & 2 the split is roughly **63% AO1 : 37% AO2** – so about a third of the marks are "apply it to something new". Don't just memorise – practise!

✱ You are GIVEN in the exam

- a **Periodic Table** (with A_r values) – use it, e.g. Cl = 35.5
- molar gas volume: 24 dm^3 at r.t.p.
- **Notes for qualitative analysis** (ion, gas & flame tests) in Papers 3 & 4 – but Paper 1 & 2 can still ask them, so learn them (Topic 12)
- a calculator is allowed in all papers

key How to read these notes

- **maroon bold** = a key word that earns the mark
- yellow highlight = learn word-for-word
- pink / blue / green = colour-coded comparisons
- yellow box = **definition**; blue box = **worked example**
- sticky notes = exam tips & traps ⚠
- little grey numbers like [2.4.3] = the syllabus point

cmd Command words

The first word of the question tells you what the examiner wants. These are the ones that catch people out:

Word	What you do
State / Give	short clear answer, no explanation needed
Define	the exact meaning (learn definitions word-for-word)
Describe	say what happens / the main features (observations!)
Explain	give the reasons *why* or *how* – use "because..." and particle ideas
Suggest	apply what you know to a new situation; several answers may be right
Deduce	work it out from the information given
Predict	say what will happen using a pattern or data
Calculate	work out a number – show working + units
Determine	get an answer using the data/graph given
Compare	similarities and/or differences – mention both things
Contrast	differences only
Identify	name / pick out / recognise
Evaluate	judge the quality / value of something (pros + cons)
Outline	main points only, briefly
Justify	back your answer with evidence
Sketch	simple drawing showing key features (correct shape)
Show (that)	structured working that ends at the given answer

Top 3 mark-losers

① no **units** or wrong sig. figs ② "it reacts" instead of an **observation** (fizzing, colour change, precipitate) ③ explaining with "atoms" when it's **ions** or **molecules**.

Preview

PREVIEW

Topic 1

States of Matter

solids, liquids, gases & what their particles get up to

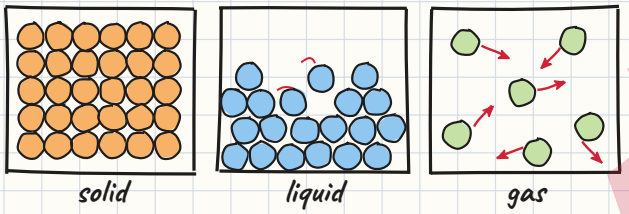
1.1 Solids, liquids & gases

✱ What you can actually see [1.1.1]

	solid	liquid	gas
shape	fixed	takes shape of container	fills the whole container
volume	fixed	fixed	not fixed
flows?	no	yes	yes
compress?	no	hardly	easily
density	high	fairly high	very low

✱ The particle model (kinetic particle theory) [1.1.2]

All matter is made of tiny **particles** (atoms, ions or molecules) that are **always moving**. How much they move depends on how much **kinetic energy** they have.



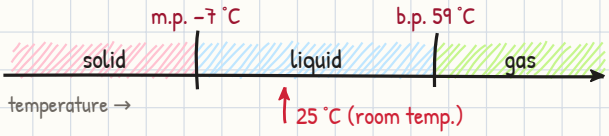
solid liquid gas

	solid	liquid	gas
separation	very close (touching)	close (mostly touching)	far apart
arrangement	regular (lattice)	irregular / random	random
motion	vibrate about fixed positions	move around, slide past each other	move fast, randomly, in all directions
attractive forces	strong	weaker	very weak
energy	lowest	more	highest

why? Gases compress easily because there is **lots of empty space** between particles. Solids & liquids can't – particles are already touching.

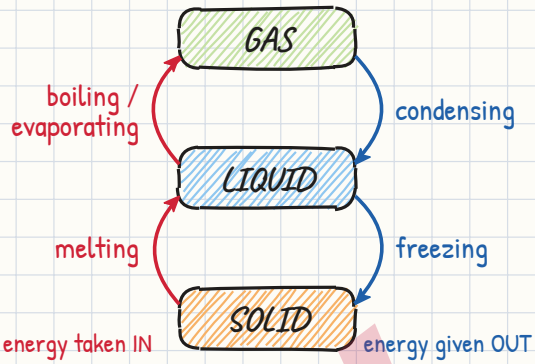
✱ Using m.p. & b.p. data to find the state

At a temperature T: T below m.p. ⟶ solid between m.p. and b.p. ⟶ liquid above b.p. ⟶ gas



e.g. bromine: m.p. −7 °C, b.p. 59 °C ⇒ at 25 °C it is a **liquid** (25 is between the two). At 70 °C it would be a gas.

✱ Changes of state [1.1.3]



- **Melting** (at the **melting point**): heating gives particles more kinetic energy ⟶ they **vibrate more** ⟶ they overcome the forces of attraction holding them in the lattice ⟶ they can move past each other.
- **Boiling** (at the **boiling point**): particles *throughout* the liquid gain enough energy to overcome the attractive forces and escape as a gas ⟶ bubbles form inside the liquid.
- **Evaporating**: happens at **any temperature below the b.p.**, only at the **surface**. The most energetic particles escape, so the liquid left behind **cools down** (that's why sweat cools you!).
- **Condensing**: a gas cools ⟶ particles lose energy, slow down ⟶ attractive forces pull them close together ⟶ liquid.
- **Freezing**: a liquid cools ⟶ particles lose energy ⟶ settle into a fixed, regular lattice. freezing point = melting point for a pure substance.

	evaporation	boiling
temperature	any temp. below b.p.	only at the b.p.
where?	surface only	throughout the liquid (bubbles)
speed	slow	fast

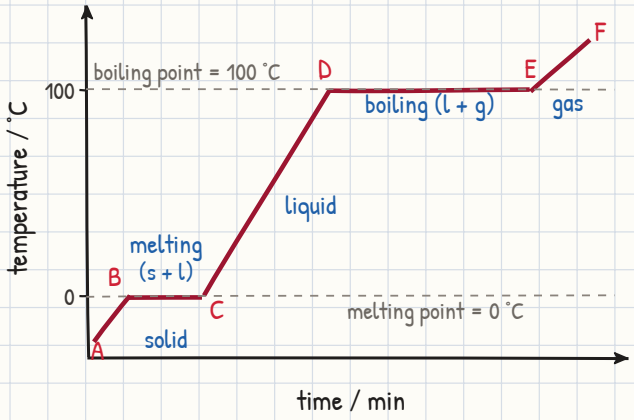
exam tip

In a "describe/explain" particle question always cover 3 **things**: arrangement, separation and motion – plus energy & forces if it's a change of state. One mark each, usually!

Preview

1.1 Heating & cooling curves

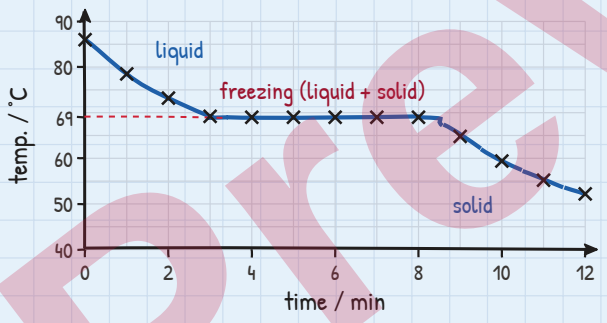
[1.1.4] – e.g. heating ice at a steady rate until it's all steam



- **Sloping parts** (AB, CD, EF): energy goes into increasing the particles' **kinetic energy** ⟶ temperature rises.
- **Flat parts** (BC, DE): temperature stays constant because the energy taken in is used to overcome the forces of attraction between particles, not to speed them up.
- DE is **longer** than BC: much more energy is needed to separate particles completely (liquid ⟶ gas) than just loosen them (solid ⟶ liquid).
- A **pure** substance gives a **sharp** flat line. An impure one melts over a **range** (and melts lower / boils higher).

Worked example ✎ cooling curve

Liquid X (molten wax) cools in a boiling tube; temperature taken every minute:

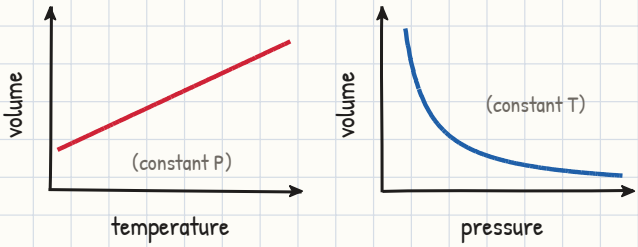


Q: What is the freezing point of X? Why is the line flat from 3 to 8 min?

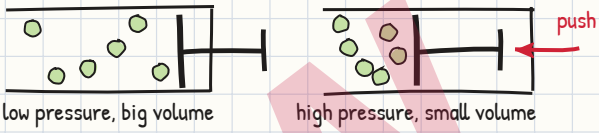
A: 69 °C – read off the flat part. While freezing, energy is **released** as attractive forces form between particles; this balances the heat lost to the surroundings, so the temperature stays constant until all of X is solid.

✱ Gases: temperature, pressure & volume [1.1.5]

Gas **pressure** = gas particles **colliding with the walls** of the container.



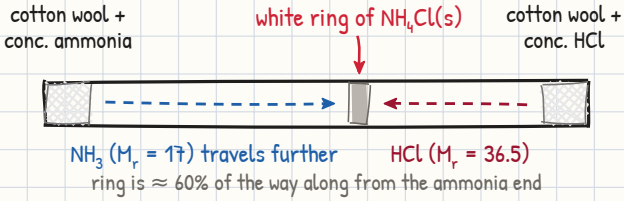
- **Temperature ↑** (pressure kept constant): particles gain kinetic energy, **move faster** and hit the walls harder and more often ⟶ the gas expands, volume increases.
- **Pressure ↑** (temperature kept constant): the particles are pushed **closer together** ⟶ volume decreases. Double the pressure ≈ half the volume.



1.2 Diffusion

Diffusion: the net movement of particles from a region of higher concentration to a region of lower concentration, caused by the **random motion** of the particles. [1.2.1]

- Fast in **gases**, slower in liquids (particles move slower, less space), basically none in solids.
- **Higher temperature** ⟶ particles move faster ⟶ faster diffusion.
- Lower M_r ⟶ faster diffusion. At the same temperature lighter molecules move faster. [1.2.2]



$$NH_3(g) + HCl(g) \longrightarrow NH_4Cl(s)$$

Worked example ✎

Q: Explain why the white ring forms nearer the HCl end. [3]

A: ① both gases **diffuse** along the tube by random motion; ② NH_3 has a **lower M_r** (17) than HCl (36.5), so its molecules **move faster**; ③ NH_3 travels **further** in the same time, so they meet closer to the HCl end.

Preview

Topic 1 · let's test it
Practice Questions

PAPER 1 · MULTIPLE CHOICE

circle one answer – A, B, C or D

1 Which describes the particles in a **liquid**?

A far apart, random arrangement, move quickly
B close together, regular arrangement, vibrate only
C far apart, regular arrangement, vibrate only
D close together, random arrangement, move past each other

2 Substance Q melts at −7 °C and boils at 59 °C. What is its state at 25 °C?

A solid **B** liquid **C** gas **D** s + l

3 Which change of state **releases** energy to the surroundings?

A boiling **B** evaporating **C** melting **D** freezing

4 While a pure solid is melting, its temperature stays constant. Why?

A the particles stop moving
B the kinetic energy of the particles is increasing
C energy is being used to overcome the forces between particles
D no energy is being taken in

5 Which gas diffuses **fastest** at room temperature?

A CO_2 **B** O_2 **C** N_2 **D** CH_4

6 A gas is heated in a syringe at constant pressure. What happens?

A the particles move faster and the volume increases
B the particles get bigger and the volume increases
C the number of particles increases
D the particles move slower and the volume decreases

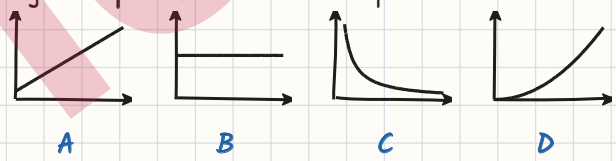
7 Which statement about evaporation is correct?

A it happens throughout the liquid
B it can happen below the boiling point
C it only happens at the boiling point
D it makes the remaining liquid warmer

8 Which property do gases have but liquids do **not**?

A they flow
B they take the shape of their container
C they have a fixed volume
D they can be compressed easily

9 Which graph shows how the volume of a fixed mass of gas changes with **pressure** at constant temperature?



10 In the NH_3/HCl diffusion tube, why does the white solid form nearer the HCl end?

A HCl molecules have a higher M_r so diffuse more slowly
B HCl molecules have a lower M_r so diffuse more slowly
C ammonia is denser than hydrogen chloride
D ammonia molecules are larger

PAPER 2 · STRUCTURED

1 Ammonia and hydrogen chloride gases diffuse towards each other along a glass tube .

(a) Define *diffusion*. [2]

(b) (i) Name the white solid formed. [1]
(ii) Write the symbol equation, with state symbols, for its formation. [2]
(iii) Explain why the white solid forms closer to the HCl end. [3]

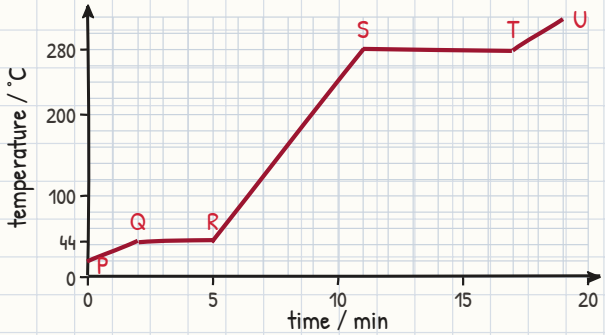
(c) The experiment is repeated at a higher temperature. State and explain the effect on the time taken for the white solid to appear. [2]

(d) HCl is replaced by hydrogen bromide, HBr. Predict whether the white solid forms nearer to, or further from, the ammonia end than before. Explain. [2]

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2 Solid Z is heated at a steady rate. The graph shows its temperature.



(a) State the melting point of Z. [1]

(b) State the physical state(s) of Z between Q and R. [1]

(c) Explain why the temperature does not change between Q and R. [2]

(d) Describe the arrangement and motion of the particles of Z between R and S. [2]

(e) Suggest why the section ST is longer than QR. [1]

(f) A sample of Z contains an impurity. How would its melting point differ? [1]

3 The table gives data for four substances.

substance	m.p. / °C	b.p. / °C
A	−218	−183
B	801	1413
C	−39	357
D	17	118

(a) Which substance is a gas at 25 °C? [1]

(b) Which substance is a liquid over the largest range of temperature? Show how you decided. [2]

(c) State the physical state of D at (i) 10 °C, (ii) 25 °C. [2]

(d) All four substances are gases at 1500 °C and are then cooled. Which one condenses first? [1]

(e) Explain, in terms of particles, why a gas can be compressed easily but a solid cannot. [2]

4 A student leaves a beaker of ethanol on the bench. The volume decreases over several hours, even though the room is at 22 °C and ethanol boils at 78 °C.

(a) Name the process taking place. [1]

(b) Explain, in terms of particles, how ethanol can turn into a gas below its boiling point. [2]

(c) Suggest two changes that would make the ethanol disappear faster. [2]

(d) The temperature of the ethanol left in the beaker is slightly lower than room temperature. Suggest why. [1]

Quick-fire recall ✦ *cover the notes & say these out loud · Topic 1 total: P1 = 10, P2 = 32 marks · answers from pg. 82*

1. 3 differences between particles in a solid and a gas
2. Why does temperature stay constant while boiling?
3. 2 differences between boiling and evaporation
4. What causes gas pressure?
5. Define diffusion (word-for-word!)
6. Effect of M_r on rate of diffusion
7. How does an impurity change m.p. and b.p.?
8. Which changes of state are exothermic?

Preview

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Topic 2

Atoms, Elements & Compounds

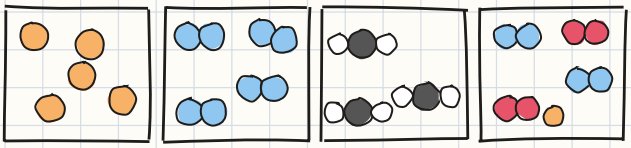
what stuff is made of & how it holds together

2.1 Elements, compounds, mixtures

Element: a substance made of only **one type of atom**; it can't be broken down into anything simpler by chemical reactions.

Compound: two or more **different elements chemically combined** (bonded) in a **fixed ratio**.

Mixture: two or more substances **not chemically combined** – can be separated by **physical** methods (filtering, distilling…).



element (atoms, e.g. Fe) | element (molecules, e.g. O_2) | compound (e.g. CO_2) | mixture (e.g. air)

	compound	mixture
make-up	fixed ratio (formula)	any proportions
properties	new – different from its elements	keeps properties of each part
separate by	chemical reaction only	physical methods
forming it	energy change usually seen	usually none
melting	sharp m.p.	over a range

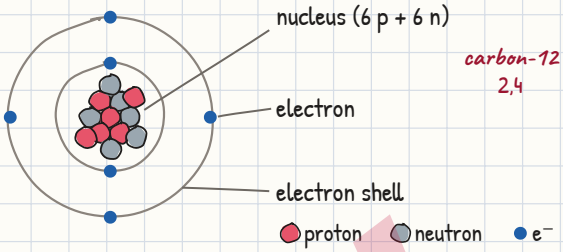
classic example: iron + sulfur

Mixture of iron filings & sulfur: a magnet pulls the iron out. Heat it ⟶ it glows and forms **iron(II) sulfide**, FeS – a **compound**: not magnetic, can't be separated physically.

remember: a **pure substance** is a single element *or* a single compound. Air, sea water, crude oil, alloys ⇒ all **mixtures**.

2.2 Atomic structure

An atom = a tiny central **nucleus** (protons + neutrons) surrounded by **electrons** in **shells**. [2.2.1]



carbon-12
2,4

particle	relative charge	relative mass	where?
proton	+1	1	nucleus
neutron	0	1	nucleus
electron	−1	$^1/_{1840}$ (tiny!)	shells

Proton number / atomic number (Z): the number of **protons** in the nucleus of an atom.

Mass number / nucleon number (A): the **total number of protons and neutrons** in the nucleus.

mass number (p + n) → 23
proton number (also = e^- in atom) → 11
Na

protons = 11
neutrons = 23 − 11 = 12
electrons = 11

Worked example ✎ ions too [2.3.3]

For an ion, **electrons = protons − charge**.

$^{27}_{13}Al^{3+}$: p = 13, n = 27 − 13 = **14**, e = 13 − 3 = **10**

$^{35}_{17}Cl^-$: p = 17, n = 35 − 17 = **18**, e = 17 + 1 = **18**

your turn ✎

How many protons, neutrons & electrons in $^{40}Ca^{2+}$ (Z = 20) and $^{16}O^{2-}$ (Z = 8)?

(answers: 20, 20, 18 | 8, 8, 10)

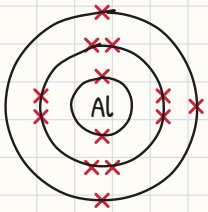
Almost all the **mass** of an atom is in the tiny nucleus (electrons weigh almost nothing). Atoms are **neutral**: protons = electrons.

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2.2 Electronic configuration

Electrons fill shells from the **inside out**. For the first 20 elements: 1st shell max 2, 2nd max 8, 3rd shell 8, then the 4th starts. [2.2.5]

Z		config	Z		config
1	H	1	11	Na	2,8,1
2	He	2	12	Mg	2,8,2
3	Li	2,1	13	Al	2,8,3
4	Be	2,2	14	Si	2,8,4
5	B	2,3	15	P	2,8,5
6	C	2,4	16	S	2,8,6
7	N	2,5	17	Cl	2,8,7
8	O	2,6	18	Ar	2,8,8
9	F	2,7	19	K	2,8,8,1
10	Ne	2,8	20	Ca	2,8,8,2



aluminium, Z = 13 ⟶ **2,8,3**
3 occupied shells ⟶ Period 3
3 outer electrons ⟶ Group III

✱ Patterns you must state [2.2.6]

- Group VIII noble gases have a full outer shell (He = 2, Ne = 2,8, Ar = 2,8,8) ⟶ unreactive.
- Groups I–VII: **number of outer-shell electrons = group number.**
- **number of occupied shells = period number.**

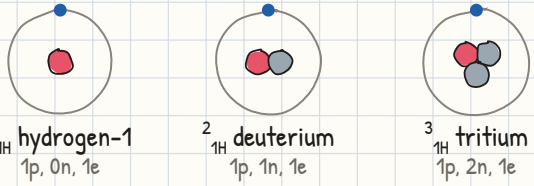
✱ Ions have configurations too

Atoms lose/gain electrons to get a **noble gas configuration**:

atom	⟶	ion	same as
Na 2,8,1	lose $1e^-$	Na^+ 2,8	Ne
Mg 2,8,2	lose $2e^-$	Mg^{2+} 2,8	Ne
O 2,6	gain $2e^-$	O^{2-} 2,8	Ne
Cl 2,8,7	gain $1e^-$	Cl^- 2,8,8	Ar
Ca 2,8,8,2	lose $2e^-$	Ca^{2+} 2,8,8	Ar

2.3 Isotopes

Isotopes: different atoms of the same element that have the same number of protons but different numbers of neutrons.



- Isotopes have the **same chemical properties** because they have the **same number of electrons** and so the **same electronic configuration** — and reactions only involve electrons. [2.3.2]
- They differ only in **mass** (so slightly different density, rate of diffusion…).

✱ Relative atomic mass from isotopes [2.3.4]

$$A_r = \frac{\Sigma\ (\text{isotope mass} \times \%\ \text{abundance})}{100}$$

Worked example 1 ✎ chlorine

75% ^{35}Cl and 25% ^{37}Cl

$A_r = (35 \times 75 + 37 \times 25) \div 100 = (2625 + 925) \div 100 =$ **35.5**

…that's why the Periodic Table says Cl = 35.5 — it's an **average**, not a real atom!

Worked example 2 ✎ three isotopes

Magnesium: 79% ^{24}Mg, 10% ^{25}Mg, 11% ^{26}Mg

$A_r = (24\times79 + 25\times10 + 26\times11) \div 100 = (1896 + 250 + 286) \div 100 = 24.32 \approx$ **24.3**

Worked example 3 ✎ backwards!

Bromine (A_r = 80) is ^{79}Br and ^{81}Br. Find the % of ^{79}Br.

Let ^{79}Br = x %, so ^{81}Br = (100 − x) %

$79x + 81(100 - x) = 80 \times 100 \Rightarrow 8100 - 2x = 8000 \Rightarrow x =$ **50%**

sanity check ✓

Your A_r must lie **between** the lightest and heaviest isotope, and closer to the most abundant one.

Preview

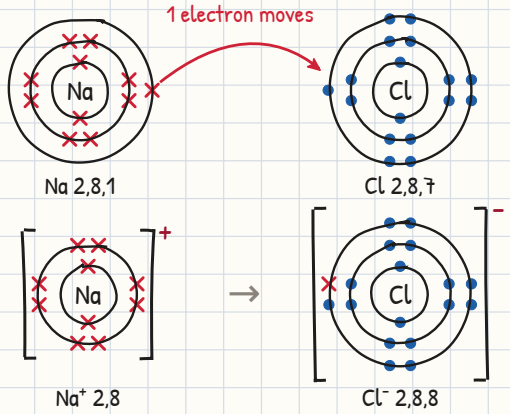
PREVIEW

2.4 Ions & ionic bonds

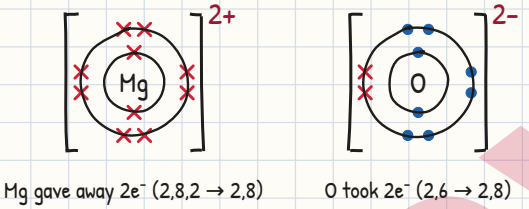
- **Cation** = positive ion. Metals **lose** outer electrons:
  $Na \longrightarrow Na^+ + e^-$, $Mg \longrightarrow Mg^{2+} + 2e^-$
- **Anion** = negative ion. Non-metals **gain** electrons:
  $Cl + e^- \longrightarrow Cl^-$, $O + 2e^- \longrightarrow O^{2-}$

Ionic bond: a strong electrostatic attraction between oppositely charged ions. [2.4.3]

✱ Dot-and-cross: sodium chloride [2.4.4]



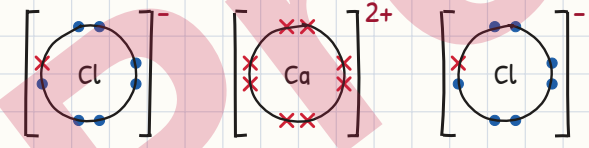
✱ Magnesium oxide, MgO



Mg gave away $2e^-$ (2,8,2 → 2,8) O took $2e^-$ (2,6 → 2,8)

key: × = electrons from the metal, ● = electrons from the non-metal. Show the **charge** and the **brackets**!

✱ Calcium chloride, $CaCl_2$ (outer shells only)

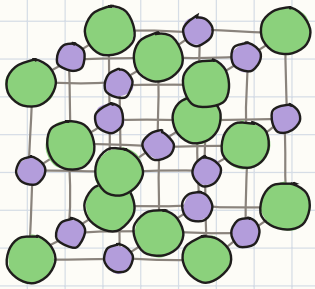


Ca (2,8,8,2) gives one e^- to EACH Cl atom

How to draw any ionic dot-and-cross

1. write both electronic configurations
2. metal **loses** its outer electrons (they go to the non-metal)
3. non-metal now has a **full outer shell** – draw the gained electron(s) with the *other* symbol
4. square **brackets** round each ion + the **charge** top right
5. check the charges cancel (e.g. Ca^{2+} needs **two** Cl^-)

✱ Giant ionic lattice [2.4.2]



A **regular** arrangement of **alternating** positive and negative ions, repeated in all 3 directions. Each Na^+ is surrounded by 6 Cl^- (and vice versa).

✱ Properties of ionic compounds [2.4.5]

property	*why?*
high melting & boiling points	lots of energy needed to overcome the strong electrostatic attractions between the oppositely charged ions, acting in all directions throughout the lattice
don't conduct when solid	ions are held in fixed positions – they **can't move**
conduct when molten or aqueous	the lattice breaks up, so the ions are free to move and carry the charge

⚠ *the #1 trap*

In molten/aqueous ionic compounds it's the **IONS** that move, not electrons. Write "electrons move" ⟶ zero marks.

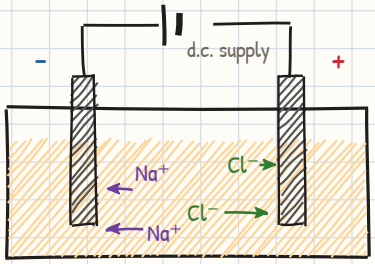
✱ Formulas of ionic compounds

The charges must **cancel out** (overall charge = 0):

ions	*balance*	*formula*
Na^+ Cl^-	1 : 1	$NaCl$
Ca^{2+} Cl^-	1 : 2	$CaCl_2$
Na^+ O^{2-}	2 : 1	Na_2O
Al^{3+} O^{2-}	2 : 3	Al_2O_3

More on formulas & the "swap and drop" trick in Topic 3 ⟶

✱ Seeing it: molten NaCl conducts



Na^+ ions move to the − electrode, Cl^- ions move to the + electrode – the ions carry the current. (full story in Topic 4)

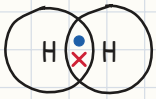
Preview

PREVIEW

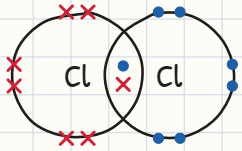
2.5 Simple molecules & covalent bonds

Covalent bond: formed when a pair of electrons is shared between two atoms, leading to **noble gas electronic configurations**. Usually between **non-metal** atoms. [2.5.1]

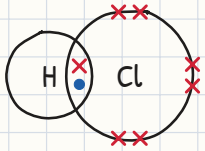
Dot-and-cross diagrams (outer shells only). Count the electrons around each atom: **H = 2**, everything else = **8**. [2.5.2] × and ● show which atom each electron came from (in CH_3OH, ● = oxygen's).



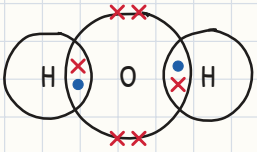
hydrogen, H_2 (H–H)



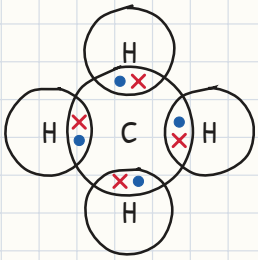
chlorine, Cl_2 (Cl–Cl)



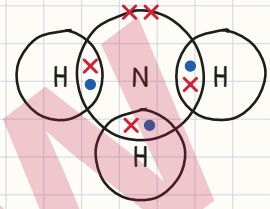
hydrogen chloride, HCl



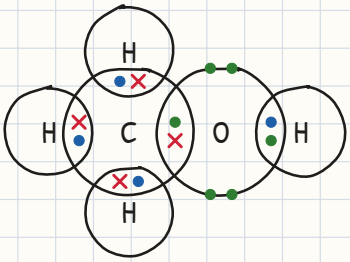
water, H_2O



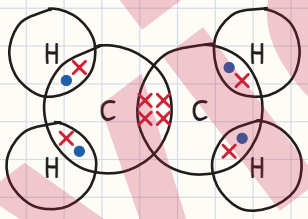
methane, CH_4



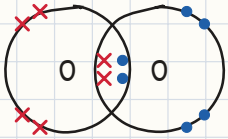
ammonia, NH_3



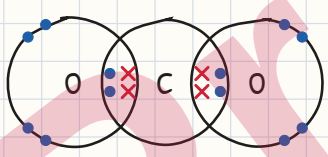
methanol, CH_3OH



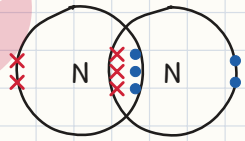
ethene, C_2H_4 (C=C)



oxygen, O_2 (O=O)



carbon dioxide, CO_2 (O=C=O)

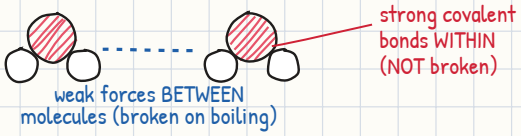


nitrogen, N_2 (N≡N)

count to check ✓
single bond = **1** shared pair (2 e⁻)
double bond = **2** shared pairs (4 e⁻)
triple bond = **3** shared pairs (6 e⁻)
Lone pairs are **not** shared — don't forget them!

✱ Properties of simple molecular substances [2.5.3]

- **Low melting & boiling points** (many are gases or liquids at r.t.p.) — only the weak intermolecular forces between molecules need to be overcome.
- **Poor electrical conductivity** — no ions and no free/delocalised electrons to carry charge.



trap! Boiling water does **not** break O–H bonds — steam is still H_2O molecules.

How to draw a covalent dot-and-cross

1. work out outer electrons: H 1, C 4, N 5, O 6, Cl 7
2. each atom needs (8 − outer e⁻) bonds: C 4, N 3, O 2, Cl 1, H 1
3. overlap the circles; put each **shared pair** in the overlap (one × + one ●)
4. add the **lone pairs** so every atom has 8 (H: 2)
5. count again ✓

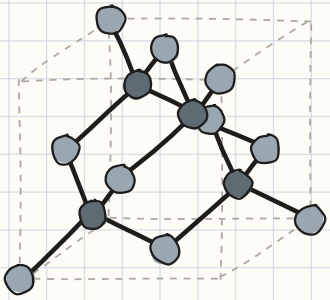
molecule	bonds	bonding pairs	lone pairs (central atom)
CH_4	4 single	4	0
NH_3	3 single	3	1
H_2O	2 single	2	2
CO_2	2 double	4	0 (each O has 2)
N_2	1 triple	3	1 on each N

preview

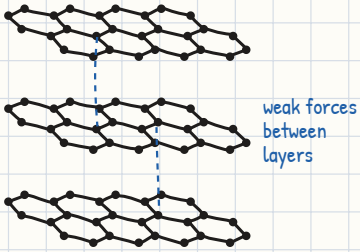
PREVIEW

2.6 Giant covalent structures

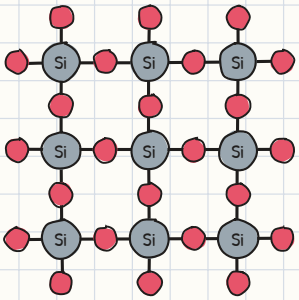
Billions of atoms joined by **strong covalent bonds** in a continuous 3-D network — no small molecules. [2.6.1]



diamond (C)



graphite (C)

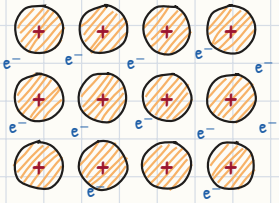


silicon(IV) oxide, SiO_2

	diamond	graphite	silicon(IV) oxide, SiO_2
structure	each C bonded to **4** other C atoms (tetrahedral)	each C bonded to **3** others in flat **hexagonal layers**; 4th electron of each C is delocalised	each Si bonded to **4** O; each O bonded to **2** Si (ratio 1 : 2)
properties	very hard, very high m.p., no conduction (no free electrons)	soft & slippery; conducts electricity; very high m.p.	hard, very high m.p., no conduction — **like diamond**
use & why	**cutting tools** — hardest natural substance because of strong covalent bonds in all directions	**lubricant** — weak forces between layers so layers slide; **electrode** — delocalised electrons carry charge, high m.p.	(sand, glass making — not needed)

why so similar? Diamond and SiO_2 both have a **giant 3-D network** where every atom is held by **strong covalent bonds** in every direction ⟶ both hard, both very high m.p., neither conducts. [2.6.3]

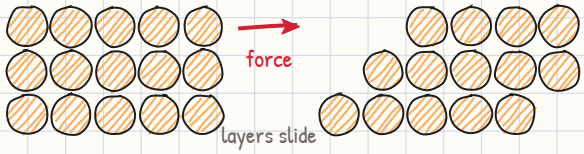
2.7 Metallic bonding



positive metal ions in a giant lattice + a **"sea" of delocalised electrons** moving freely between them

Metallic bonding: the electrostatic attraction between the positive ions in a giant metallic lattice and a 'sea' of delocalised electrons.

- **Good electrical conductors**: delocalised electrons are free to **move through the structure** and carry charge.
- **Malleable & ductile**: the layers of positive ions can **slide over each other** without breaking the bonding (the electron sea just moves with them).



✱ Summary — which structure is it?

	particles	m.p./b.p.	conducts?
giant ionic NaCl, MgO	ions	high	only molten / aqueous
simple molecular H_2O, CO_2, CH_4	small molecules	**low**	no
giant covalent diamond, SiO_2	atoms (network)	very high	no (graphite: yes)
metallic Cu, Fe, Na	+ ions & delocalised e^-	usually high	**yes**, solid & liquid

exam detective

High m.p. + conducts only when molten ⇒ **ionic**. Low m.p. + never conducts ⇒ **simple molecular**. Very high m.p. + never conducts ⇒ **giant covalent**. Conducts as a solid ⇒ **metal** (or graphite).

Model answers (learn the wording!)

Q Why does diamond have a very high m.p.? ⟶ *giant covalent structure; many **strong covalent bonds** must be broken, which needs a lot of energy.*

Q Why does graphite conduct? ⟶ *each C has one **delocalised electron** which can move through the structure.*

Q Why is CO_2 a gas? ⟶ *simple molecules; **weak intermolecular forces** need little energy to overcome.*

Preview



Topic 2 · let's test it

Practice Questions



PAPER 1 · MULTIPLE CHOICE

1 Which particle has a relative charge of −1 and a negligible mass?

A electron B neutron C proton D nucleus

2 An atom of element X has 19 protons and 20 neutrons. Which statement is correct?

A the mass number of X is 19
B the proton number of X is 39
C X is in Group 1 and Period 4
D an atom of X has 20 electrons

3 How many electrons are in the ion $_{16}^{32}S^{2-}$?

A 14 B 16 C 32 D 18

4 Atoms of two isotopes of the same element have the same

A mass number B number of neutrons
C number of electrons D physical properties

5 Element Y is 60% ^{69}Y and 40% ^{71}Y. What is the relative atomic mass of Y?

A 69.4 B 69.8 C 70.0 D 70.2

6 Which pair of elements forms an **ionic** compound?

A carbon and oxygen B hydrogen and chlorine
C magnesium and fluorine D nitrogen and hydrogen

7 How many lone (non-bonding) pairs of electrons are on the nitrogen atom in NH_3?

A 0 B 1 C 2 D 3

8 Which substance conducts electricity both as a solid and when molten?

A sodium chloride B copper
C sulfur D silicon(IV) oxide

9 Why is graphite used as a lubricant?

A it has delocalised electrons
B each carbon atom forms four covalent bonds
C it has a very high melting point
D there are weak forces between its layers, so they slide

10 Substance Z melts at 1610 °C and does not conduct electricity when solid or molten. What type of structure does Z have?

A giant covalent B giant ionic
C metallic D simple molecular

11 Which molecule contains a **triple** covalent bond?

A O_2 B CO_2 C C_2H_4 D N_2

12 Which statement about ionic compounds is correct?

A they conduct when molten because the ions can move
B they conduct when solid because the ions can move
C they have low melting points
D they conduct because electrons move through the lattice

PAPER 2 · STRUCTURED

1 Magnesium has isotopes including $_{12}^{24}Mg$ and $_{12}^{26}Mg$.

(a) Complete the table. [3]

	atom of ^{24}Mg	ion $^{26}Mg^{2+}$
protons		
neutrons		
electrons		

(b) Explain why ^{24}Mg and ^{26}Mg have the same chemical properties. [2]

.....

.....

(c) A sample of magnesium is 78.6% ^{24}Mg, 10.1% ^{25}Mg and 11.3% ^{26}Mg. Calculate the relative atomic mass of magnesium to 1 decimal place. [2]

(d) Give the electronic configuration of a Mg^{2+} ion. [1]

.....

(e) Magnesium is in Group II. Explain how you could predict this from its electronic configuration. [1]

.....

Preview

PREVIEW

2 Lithium reacts with oxygen to form lithium oxide, Li_2O.

(a) Draw a dot-and-cross diagram for lithium oxide. Show the outer shells only and the charges on the ions. [3]

draw here

(b) Explain why lithium oxide has a high melting point. [2]

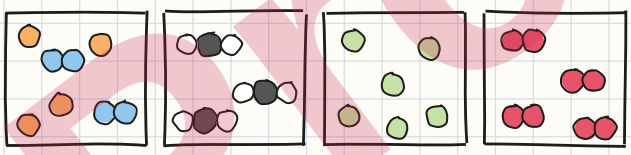
(c) Explain why solid lithium oxide does not conduct electricity, but molten lithium oxide does. [2]

(d) Nitrogen trichloride, NCl_3, is a covalent compound. Draw a dot-and-cross diagram for one molecule of NCl_3, showing outer electrons only. [2]

draw here

(e) Explain why NCl_3 has a low boiling point. [2]

3 The diagrams show the particles in four substances. (each circle = an atom; different shades = different elements)



(a) Which diagram shows (i) a compound, (ii) a mixture of two elements, (iii) an element made of single atoms, (iv) an element made of diatomic molecules? [4]

(b) State **two** differences between a compound and a mixture. [2]

4 The table gives information about four solids.

solid	m.p. / °C	conducts as a solid?	conducts when molten?
P	801	no	yes
Q	−114	no	no
R	1083	yes	yes
S	1710	no	no

(a) Which solid is (i) a metal, (ii) a simple molecular substance, (iii) a giant ionic substance, (iv) a giant covalent substance? [4]

(b) Describe the bonding in R. [2]

(c) Explain why R is malleable. [2]

(d) S is silicon(IV) oxide. Describe its structure. [2]

(e) Graphite and diamond are both pure carbon. Explain why graphite conducts electricity but diamond does not. [2]

mark yourself!
Paper 1 = 12 marks. Paper 2 = 9 + 11 + 6 + 12 = **38 marks**. Answers from page 82. Got a dot-and-cross wrong? Redraw it 3 times — seriously, it sticks.

Quick-fire recall ✦

1. Define isotopes & say why they react the same.
2. Define ionic bond, covalent bond, metallic bonding.
3. Draw CO_2, N_2 and CH_3OH from memory.
4. Why does graphite conduct but diamond doesn't?
5. Why don't solid ionic compounds conduct?

Preview

PREVIEW

Topic 3

Stoichiometry

formulas, equations & the mighty mole (a.k.a. chemistry maths)

3.1 Formulas

✱ Formulas of elements [3.1.1]

Most elements are just their symbol: Fe, Na, C, Mg, He, S… but **seven** exist as **diatomic molecules**:

$H_2\ N_2\ O_2\ F_2\ Cl_2\ Br_2\ I_2$

"Have No Fear Of Ice Cold Beverages"

(H, N, F, O, I, Cl, Br)

✱ Ion cheat sheet — learn these charges!

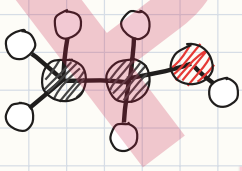
charge	positive ions	negative ions
1	H^+, Li^+, Na^+, K^+, Ag^+, NH_4^+	F^-, Cl^-, Br^-, I^-, OH^-, NO_3^-
2	Mg^{2+}, Ca^{2+}, Ba^{2+}, Cu^{2+}, Zn^{2+}, Fe^{2+}, Pb^{2+}	O^{2-}, S^{2-}, SO_4^{2-}, CO_3^{2-}, SO_3^{2-}
3	Al^{3+}, Fe^{3+}, Cr^{3+}	(N^{3-})

blue = groups of atoms that stay together (compound ions). The Roman numeral tells you the charge: iron(III) = Fe^{3+}, copper(II) = Cu^{2+}.

Molecular formula: the **number and type of different atoms in one molecule.** [3.1.2]

Empirical formula: the **simplest whole number ratio** of the different atoms or ions in a compound. [3.1.3]

compound	molecular	empirical
ethene	C_2H_4	CH_2
glucose	$C_6H_{12}O_6$	CH_2O
butane	C_4H_{10}	C_2H_5
water	H_2O	H_2O (same!)



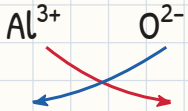
Deduce from a model [3.1.4]: just count! 2 black C, 6 white H, 1 red O ⇒ C_2H_6O (ethanol)

Formulas to just KNOW

water H_2O · carbon dioxide CO_2 · ammonia NH_3 · methane CH_4 · hydrochloric acid HCl · sulfuric acid H_2SO_4 · nitric acid HNO_3 · sodium hydroxide NaOH · calcium carbonate $CaCO_3$ · sulfur dioxide SO_2

✱ Ionic formulas from charges [3.1.5]

Total charge must be **zero**. Quick trick: "swap & drop" the charge numbers, then simplify.



→ Al_2O_3

check: 2(+3) + 3(−2) = 0 ✓

ions	formula	note
Ca^{2+} + OH^-	$Ca(OH)_2$	brackets round a compound ion if you need 2+ of it
NH_4^+ + SO_4^{2-}	$(NH_4)_2SO_4$	ammonium sulfate
Fe^{3+} + SO_4^{2-}	$Fe_2(SO_4)_3$	iron(III) sulfate
Mg^{2+} + O^{2-}	MgO	Mg_2O_2 ⟶ simplify!
Cu^{2+} + NO_3^-	$Cu(NO_3)_2$	copper(II) nitrate

✱ Equations [3.1.6–7]

word ⟶ **symbol** ⟶ **balance** (change the big numbers in front, never the little ones) ⟶ add **state symbols**: (s) solid, (l) liquid, (g) gas, (aq) dissolved in water.

Worked example ✎ balancing

magnesium + oxygen ⟶ magnesium oxide

$Mg + O_2 \longrightarrow MgO$ (2 O on left, 1 on right ✗)

$Mg + O_2 \longrightarrow 2MgO$ (now Mg is wrong ✗)

$2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$ ✓

more: $CH_4 + 2O_2 \longrightarrow CO_2 + 2H_2O$ | $Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$

✱ Ionic equations

Show **only the particles that change**. Split **aqueous** ionic compounds into ions, then cross out the **spectator ions** (same on both sides).

Worked example ✎

$AgNO_3(aq) + NaCl(aq) \longrightarrow AgCl(s) + NaNO_3(aq)$

$Ag^+ + \cancel{NO_3^-} + \cancel{Na^+} + Cl^- \longrightarrow AgCl + \cancel{Na^+} + \cancel{NO_3^-}$

$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$

Learn these too: $H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$ (neutralisation)

$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$ (displacement)

Preview